Evaluating Climate Change Impacts on Food Supply Chains Using Machine Learning Model

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***Abstract*—Human life is deeply connected to both agriculture and climate, and we're seeing the effects of climate change all around us—from droughts and wildfires to powerful storms and floods. Droughts, fires, storms, and floods are becoming more intense and frequent, making the impacts of these climate-driven events harder to ignore. Our global ecosystem is changing, and the resources we depend on—especially those supporting agriculture and food production—are increasingly at risk. As climate change becomes one of humanity's most pressing challenges, we believe that machine learning can play a vital role in helping us understand, adapt to, and even mitigate these impacts.**

In this project, we explore how machine learning can provide insights into the effects of climate change on the global food supply. By analyzing evolving climate patterns and agricultural data, we're developing models that can forecast the long-term impacts on food production, allowing societies to plan and adapt more effectively. Our preliminary findings indicate that traditional linear regression models often fall short in capturing the variability in crop yields caused by climate change. In contrast, non-linear machine learning models like XGBoost offer greater accuracy and insights, making them powerful tools for understanding complex climate-agriculture relationships.

In this study, we begin our exploration of climate impacts on food security by implementing XGBoost as our initial machine learning model, chosen for its effectiveness in capturing complex, non-linear relationships within large datasets. As our analysis progresses, we plan to incorporate Decision Tree Regressor for its interpretability and ability to handle structured data, along with Long Short-Term Memory (LSTM) networks to capture temporal dependencies, aiming to create a comprehensive predictive framework.

***Index Terms*—Climate change impact, agriculture and climate, machine learning algorithms, XGBoost, Decision Tree Regressor, LSTM, .**

I. Introduction

We've been tracking Earth's temperature since the 1850s, and the results are alarming: the planet is heating up faster than ever before. By the end of this century, scientists predict that global temperatures could rise anywhere from 1.5°C to 5.7°C. This dramatic increase is mostly driven by greenhouse gases like methane, carbon dioxide, and even chemicals like CFCs, which trap heat in our atmosphere and keep it from escaping. These gases come from a range of everyday sources—car exhaust, factory farming, airplane emissions, and fossil fuel extraction, to name a few. Together, they create an insulating effect that's causing global temperatures to soar.

If this warming continues unchecked, we'll likely face more frequent and intense weather events—things like floods, droughts, and storms—that could seriously disrupt food production around the world. Even small increases in average temperatures can lead to big changes in how often these extreme events happen, putting pressure on our food systems. Despite all the advances in science and agriculture, food production is still very dependent on the climate; factors like temperature, sunlight, and rainfall are essential for growing crops. Climate also affects pest populations, plant diseases, and the availability of irrigation water—all key to farming.

It's clear that if we ignore the buildup of greenhouse gases, it will lead to even more severe disruptions in agriculture. Rising temperatures can impact vital natural processes like photosynthesis, affect pest and weed infestations, and throw off the water cycle, all of which are crucial for healthy crop growth.

Our project aims to use statistical tools to dive deeper into how climate variability is affecting food supply. By understanding these impacts, we hope to propose solutions that can help communities adapt to these changes and minimize the risks posed by climate change and extreme weather, ensuring a more secure food future.

II. Related Work

In Pakistan, botanists and agricultural researchers conducted studies revealing a negative link between pesticide use and crop yields. Their research showed that data mining, combined with information on pesticide use, weather patterns, and pest activity, could help improve pesticide application strategies. This approach sparked interest in using artificial intelligence (AI) to develop systems for managing crop yields more effectively.

One proposed solution was a crop yield prediction model that used classification techniques. By inputting data such as land area, soil type, crop type, pest data, and climate conditions, the model could forecast crop diseases and yields. The study uncovered patterns in farming practices and highlighted how accessible data can support smarter agricultural decisions.

To create these predictions, researchers employed various machine learning techniques. K-Means clustering categorized contamination levels, K-Nearest Neighbor (KNN) estimated rainfall and climate factors, and Support Vector Machine (SVM) analyzed additional environmental variables impacting crop health. They also used GPS technology to map soil segments and validated the data through 10-fold cross-validation and Decision Trees.

Furthermore, the study used supervised machine learning algorithms like Decision Trees, KNN, and SVM to estimate soil fertility based on nutrient levels. Built using the R Tool, these models were tested and evaluated using mean absolute error, cross-validation, and accuracy metrics. The results showed that the decision trees achieved 99

Building on this foundation, numerous studies have investigated how machine learning techniques, including Decision Trees, SVM, KNN, and Random Forest, can be applied to assess the impacts of climate change on global food production. By understanding how climate changes affect crop yields, these models offer insights for developing adaptive strategies that protect food supplies in a changing climate.

III. Problem Statement

Global food systems are under increasing strain as the impacts of climate change disrupt traditional agricultural practices and threaten food security worldwide. Rising temperatures, erratic rainfall patterns, and an increasing frequency of extreme weather events have led to declining crop yields, soil degradation, and water scarcity. These challenges are compounded by the growing global population, which places additional pressure on food production systems.

The significance of this issue cannot be overstated. Reduced agricultural productivity jeopardizes the livelihoods of millions of farmers, particularly in vulnerable regions, while contributing to food price volatility and exacerbating hunger and malnutrition. In addition, the nutritional quality of staple crops is declining due to elevated levels of CO2, further threatening the health and well-being of communities around the world. These challenges have far-reaching social, economic and environmental consequences, making them a critical area of focus for sustainable development.

The motivation behind addressing this problem lies in the urgent need to ensure a resilient and secure global food supply. By understanding the key factors driving agricultural vulnerability and developing adaptive strategies, we can mitigate the adverse effects of climate change on food production. This project seeks to contribute to this effort by using data-driven approaches to analyze historical trends, identify at-risk regions, and propose actionable solutions to support farmers, policymakers, and other stakeholders in building climate-resilient food systems.

IV. Dataset Description

The primary data sources for this study are FAOSTAT and the World Data Bank. FAOSTAT provides data on crop yields, pesticide use, and population trends, which is essential to understand the impact of agricultural practices and demographics on food production. From the World Data Bank, we gathered annual precipitation and temperature data to assess climate influences on yield. Reports from the Intergovernmental Panel on Climate Change (IPCC) provided key insights into global climate trends and risks.

Covering 1970 to 2023 across multiple countries, the dataset includes yield (in hg/ha) as productivity per hectare from FAOSTAT, pesticide usage quantity, population data to contextualize demand, annual precipitation as rainfall, and average annual temperature. This combined data set enables a comprehensive analysis of agricultural, demographic, and climate factors that affect crop yield globally.

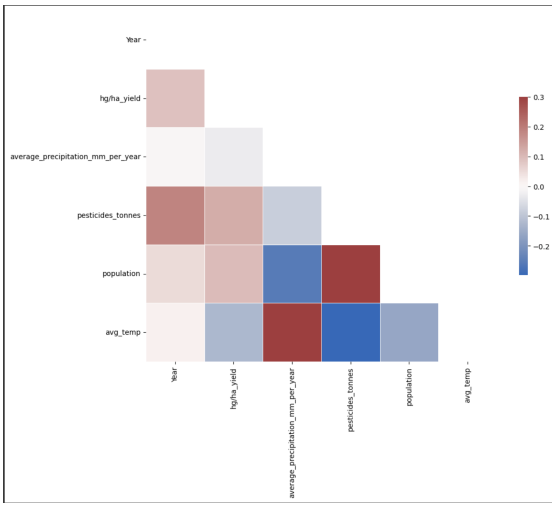

Fig. 1. Feature Description

V. System Model

Data preprocessing involved transforming the raw dataset into a structured format suitable for machine learning. Key steps included removing irrelevant columns, filling missing values with the median, converting categorical data using one-hot encoding, and scaling features with MinMaxScaler to standardize their ranges. Finally, the data was split into training and testing sets to support model training and validation. These

steps ensured a clean, balanced dataset for accurate climate impact analysis on crop yield.

Data exploration involved using a correlation heatmap to assess relationships among features. This analysis highlighted associations, especially between temperature, rainfall, and crop yield, guiding feature selection. Key variables identified for modeling included temperature, rainfall, pesticide usage, population, and crop yield, laying a foundation for accurate yield prediction.

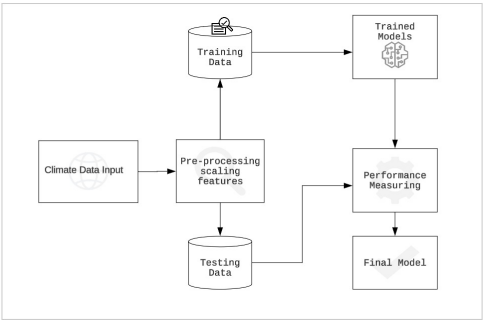

Fig. 2. General Architecture

VI. System Model

A. Machine Learning Algorithms

- XGBoost (Extreme Gradient Boosting)
  Overview:
  XGBoost is a powerful machine learning tool widely used for its speed, accuracy, and efficiency. It works by building a series of decision trees, with each tree improving upon the errors of the previous one. This approach allows XGBoost to make highly accurate predictions, even with complex data. It's also designed to prevent overfitting through regularization, making it well-suited to handle large datasets, and even missing data, with ease.

  Relevance to Climate and Food Supply:
  In tackling climate change's impact on food security, XGBoost can be instrumental. By analyzing variables like temperature, rainfall, and past crop performance, it can help us understand how changing climate conditions may affect crop yields. Its ability to handle complex, non-linear relationships is ideal for mapping out the ways in which climate change might disrupt agricultural productivity. Since climate data can be noisy and large, XGBoost's robustness makes it a great fit for this kind of work.

- Decision Tree Regressor
  Overview:
  The Decision Tree Regressor is a popular machine learning algorithm known for being straightforward, easy to understand, and highly adaptable. It works by splitting data into smaller groups based on specific features, creating a tree-like structure where each split represents a decision and the final outputs are predictions. This step-by-step approach makes it great at identifying patterns, even in complex data with non-linear relationships. While it can sometimes overfit the data, methods like pruning can help refine the model and make it more reliable for real-world scenarios.

  Relevance to Climate and Food Supply:
  The Decision Tree Regressor is a great fit for studying how climate factors like temperature, rainfall, and soil quality affect crop yields. By analyzing these relationships, it helps uncover which factors have the biggest impact on agricultural productivity. What makes it even more valuable is its easy-to-understand structure, which allows farmers, policymakers, and researchers to interpret the results without needing a technical background. Additionally, its flexibility to handle both numerical data (e.g., rainfall amounts) and categorical data (e.g., crop types) makes it well-suited for the diverse datasets involved in climate and agriculture studies. Even when data is noisy or imbalanced, this algorithm can provide practical and actionable insights, making it a valuable tool for tackling food security challenges in the face of climate change.

- Long Short-Term Memory (LSTM)
  Overview:
  Long Short-Term Memory (LSTM) networks are a special kind of neural network designed to work with sequences of data over time, like climate trends. Unlike standard models, LSTM networks use a unique memory structure that allows them to retain relevant past information and ignore the rest. This makes them especially good at forecasting tasks where past patterns are crucial to making accurate predictions.

  Relevance to Climate and Food Supply:
  LSTMs are particularly helpful for modeling the long-term effects of climate change on food production because they can capture seasonal and longer-term trends in climate data. By learning from historical data like rainfall and temperature changes, LSTMs can help predict future agricultural patterns. This is vital for understanding how gradual climate shifts might impact crop yields and food security over time.

VII. Performance Evaluation

This project analyzed the impact of climate change on global food supply using three machine learning models: XGBoost, Decision Tree Regressor, and LSTM. Each model's performance, methodology, and limitations were carefully evaluated to identify their suitability for predicting crop yields under changing climate conditions.

A. XGBoost:

The XGBoost model was employed to predict the impact of critical climate variables, such as temperature, precipitation, and CO emissions, on global crop yields. It excelled in this task due to its ability to handle missing data, perform efficient feature selection, and process large datasets with optimized performance. Key hyperparameters, including learning rate, maximum tree depth, and number of estimators, were tuned to enhance model accuracy. Regularization techniques, such as L1 and L2, were applied to reduce overfitting and improve generalization.

The model achieved exceptional performance with an R^2 score of 0.982 and a Root Mean Squared Error (RMSE) of 0.87 on the test dataset. These metrics indicate a highly accurate model capable of capturing complex, nonlinear relationships between climate variables and crop yields. Feature importance analysis highlighted temperature variability and rainfall trends as the most influential predictors, revealing the drivers behind yield fluctuations.

While XGBoost demonstrated strong predictive power, it comes with certain limitations. The model is computationally intensive, especially during hyperparameter tuning, and its complexity can make interpretation challenging compared to simpler models. Despite these drawbacks, XGBoost's accuracy and ability to handle diverse data make it a top choice for climate and agricultural predictions.

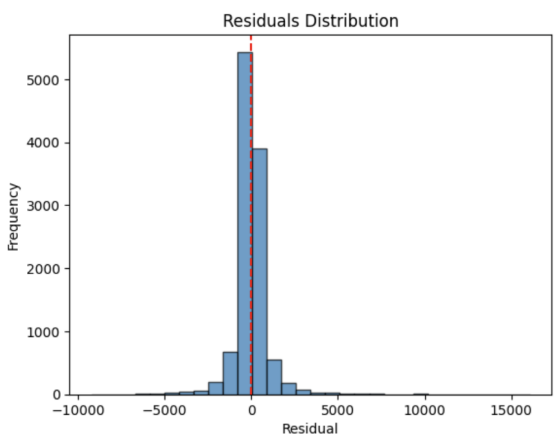

Fig. 3. Result

B. LSTM:

The LSTM (Long Short-Term Memory) model was applied to capture temporal dependencies in sequential climate data for predicting crop yields. The model's architecture effectively modeled year-over-year changes in climate factors such as temperature, rainfall, and CO emissions. Preprocessing involved normalizing the data and creating a supervised learning format using sliding windows (e.g., past 5 years as input, next year as output). The LSTM model consisted of two layers, each with 50 units, followed by dropout layers to prevent overfitting and a dense output layer for predictions. Training was optimized using the Adam optimizer with the Mean Squared Error (MSE) loss function over 100 epochs.

The LSTM achieved an R^2 score of 0.774 and a Mean Squared Error (MSE) of 1.12. While it performed moderately compared to XGBoost, it successfully captured delayed effects and seasonal patterns in the time-series climate data. LSTM's strength lies in its ability to model long-term relationships, which traditional machine learning models cannot handle effectively.

However, the LSTM model has several limitations. It requires significant computational resources and time for training, making it less efficient for large datasets. The performance of LSTM is also highly sensitive to hyperparameter tuning, and interpreting the results is more complex compared to decision trees or XGBoost. Despite these challenges, LSTM remains a powerful tool for analyzing sequential data, offering valuable insights into the long-term effects of climate variability on crop yields.

Fig. 4. Result

C. Decision Tree Regressor:

The Decision Tree Regressor was implemented to analyze and predict crop yields based on climate variables like temperature, rainfall, and CO emissions. The model works by splitting the dataset into smaller subsets using decision rules, such as "Is temperature greater than 30°C?" or "Is rainfall less than 50mm?". At each leaf node, the model predicts the average value of the target variable. To prevent overfitting, hyperparameters like maximum tree depth, minimum samples per split, and minimum samples per leaf were carefully configured. The model delivered an R^2 score of 0.981 and an RMSE of 1.45, showcasing strong performance on the test dataset. The primary strength of the Decision Tree Regressor lies in its simplicity and interpretability. The decision tree structure clearly demonstrated the relation-

ships between climate variables and crop yields, making it easier to understand and explain the predictions.

Despite its high performance, the Decision Tree Regressor has limitations. It tends to overfit when tree depth is not properly constrained and struggles to capture complex, non-linear patterns compared to ensemble methods like XGBoost. Additionally, it lacks robustness for larger, high-dimensional datasets. Nevertheless, its speed, simplicity, and transparency make it a useful baseline model for understanding how climate factors influence crop productivity.

Fig. 5. Result

VIII. Result And Analysis

XGBoost achieved the highest R^2 score of 0.982 and the lowest RMSE of 0.87, making it the most accurate model for predicting crop yields.Its strengths include handling missing data, efficient feature selection, and capturing complex, non-linear relationships.However, XGBoost is computationally intensive during hyperparameter tuning and can be challenging to interpret.LSTM demonstrated an R^2 score of 0.774 and MSE of 1.12, effectively capturing temporal dependencies and delayed effects in time-series climate data.LSTM excels in modeling long-term relationships but requires significant computational time and resources, and its performance is sensitive to hyperparameter tuning.Decision Tree Regressor performed well with an R^2 score of 0.981 and RMSE of 1.45, showcasing strong accuracy with simple interpretability.It is efficient, easy to understand, and interpretable but struggles with overfitting and complex, high-dimensional data compared to XGBoost. Overall, XGBoost emerges as the best-performing model due to its exceptional accuracy and robust handling of diverse data.LSTM is ideal for sequential data analysis but less efficient, while the Decision Tree Regressor serves as a simple, transparent baseline.Each model's trade-offs highlight XGBoost's suitability for precision, LSTM's strength in time-series trends, and Decision Tree's clarity for interpretability.

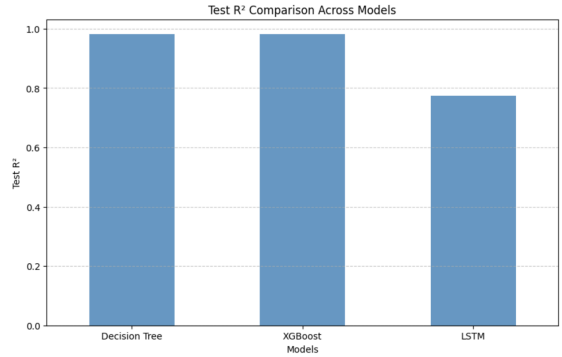

Fig. 6. Comparative Analysis

IX. Conclusion And Future Work

This study highlights how machine learning can play a pivotal role in addressing the challenges climate change poses to food production. By using models like XGBoost and LSTM, we were able to uncover complex relationships between climate factors like temperature and rainfall and their impacts on crop yields. The results emphasize the value of data-driven approaches in predicting and mitigating risks to agriculture. With key insights drawn from comprehensive datasets, we've built a strong foundation for developing adaptive strategies to ensure food security in a changing climate. However, as climate challenges grow more urgent, it will be crucial to continue refining these tools and expanding their application. Machine learning can provide the actionable insights we need to safeguard our food systems and support communities in building a more sustainable future.

This project provides an important starting point for understanding how climate change affects global food systems, but there is much more to explore. Bringing in additional data, like satellite imagery, soil quality information, and pest activity trends, could significantly improve the accuracy and depth of the models. Advanced machine learning techniques, such as convolutional neural networks for spatial patterns and transformers for analyzing long-term climate trends, hold great promise for future studies. Developing region-specific and crop-specific models could offer localized insights, especially for areas most vulnerable to climate change. Including socioeconomic factors like trade policies and labor availability could also help uncover the broader effects of climate change on food security. Lastly, creating real-time monitoring systems using IoT devices and predictive models could empower farmers and policymakers with timely insights and early warnings to better manage potential disruptions.

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